

Exploring the Role of Social Support and Community Resources in Reducing Depression among Food-Insecure Households in South Africa

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EXECUTIVE SUMMARY

Background: Food insecurity affects approximately 11.6% of households in South Africa, contributing to poor mental health outcomes, particularly depression¹. Research has consistently shown an association between food insecurity and depression, yet the role of social support and community resources in mitigating this relationship remains understudied.

Methods: This study employs a cross-sectional design, utilizing existing data sourced from Statistics South Africa's (Stats SA) General Household Survey (GHS) 2018 – 2022 series. The study employs descriptive statistics, bivariate analysis, logistic regression, and mediation analysis using Stata SE version 18. The sample includes households experiencing food insecurity, with variables capturing demographic characteristics, social support, community resources, and depression.

Outcomes: This research aims to inform evidence-based interventions addressing depression among food-insecure households in South Africa. The study's findings will inform policy development, practitioners, and researchers, leading to enhanced mental health outcomes among susceptible populations. Specifically, the findings will offer a more detailed understanding of the relationship between food insecurity, community resources, social support, and depression; identify key factors influencing depression among food-insecure households; and inform policy decisions and resource allocation for mental health services.

ABBREVIATIONS

FAO	Food and Agriculture Organization
BDI-II	Beck Depression Inventory II
CES-D	Centre for Epidemiology Studies Depression Scale
CFAP	Charitable Food Assistance Program
DMP	Data Management Plan
EA	Enumeration Area
FIES	Food Insecurity Experience Scale
GHS	General Household Survey
HFIAS	Household Food Insecurity Access Scale
MSPSS	Multidimensional Scale of Perceived Social Support
NIDS	National Income Dynamics Study
NGO	Non-Governmental Organisations
PHQ-9	Patient Health Questionnaire
PPS	Probability Proportional to Size
PSU	Primary Sampling Unit
SSQ	Social Support Questionnaire
Stats SA	Statistics South Africa

1. INTRODUCTION

Food insecurity, a pervasive issue affecting approximately 11.6% of households in South Africa, poses significant mental health risks, with depression being a significant concern¹. The complex and multifaceted relationship between food insecurity, social support, and community resources on depression is not yet fully understood, hindering the development of effective interventions. This knowledge gap necessitates research that explores the interplay between these factors to inform evidence-based policies and programs.

The consequences of food insecurity extend beyond physical health, influencing psychological well-being and mental health outcomes²⁻⁵. Depression, a leading cause of disability worldwide, is disproportionately affecting individuals experiencing food insecurity^{6, 7}. Research suggests that social support community resources may alleviate negative impacts^{8, 9}, yet their effectiveness in reducing depression among food-insecure individuals is less clearly defined.

South Africa's unique socio-economic context, characterized by high levels of poverty, inequality, and unemployment, exacerbates the food insecurity-depression relationship¹⁰. The country's social protection systems, including social grants and community-based initiatives, aim to alleviate poverty and promote well-being¹¹. However, the effectiveness of these interventions in addressing the mental health consequences of food insecurity requires further research.

This study aims to investigate the effectiveness of social support and community resources in reducing depression among individuals experiencing food insecurity in South Africa. Using secondary data from the 2018-2022 General Household Survey (GHS) conducted by Statistics South Africa (Stats SA), this research will provide valuable insights into the relationships between food insecurity, social support, community resources, and depression outcomes.

2. LITERATURE REVIEW

Food Insecurity in South Africa

Food insecurity refers to a situation where people lack reliable access to adequate food¹². In South Africa, the prevalence of food insecurity is measured using the Food Insecurity Experience Scale (FIES) as part of the General Household Survey conducted by Stats SA¹³. Research undertaken by “Wambogo EA et al.,2018” revealed that the FIES substantially satisfies the Rasch model’s criteria for equal item discrimination and conditional independence¹⁴. This validation underscores the effectiveness of FIES in assessing food insecurity in the context of South Africa.

Food insecurity remains a pervasive issue in South Africa, with far-reaching consequences for public health and economic development¹. According to the Food and Agriculture Organization (FAO), the number of people experiencing hunger worldwide has increased by 150 million since 2019, reaching 828 million in 2021¹. Africa has been disproportionately affected, with 278 million people facing hunger in 2022. In South Africa, approximately 11.6% of households (around 2.1 million) reported experiencing hunger in 2021¹. The Household Food Insecurity Access Scale (HFIAS) reveals a significant rural-urban gradient, with rural households facing higher levels of food insecurity than their peri-urban counterparts¹⁵. A 2022 report indicates a decline in South Africa's food security environment score to 61.7 points¹⁶. Furthermore, a 2020 survey by IPSOS highlights the widespread nature of hunger across the nine provinces, with KwaZulu-Natal and the Eastern Cape being the most affected, reporting 58% and 56% hunger prevalence, respectively¹⁷.

The effects of food insecurity extend beyond nutrition, impacting physical health outcomes and mental health status. Chronic malnutrition, a direct result of food insecurity, increases the risk of infection, death, and poor cognitive development in children¹⁸. Furthermore, food insecurity has been shown to hinder cognitive function in school-aged children, resulting in impaired learning capacity and attention². In South Africa, approximately half of children under five suffer from malnutrition, which accounts for 4-25% of child mortality¹⁵. The effects of food insecurity extend beyond childhood, persisting into adulthood and even affecting subsequent generations. Longitudinal studies, such as the Dutch Winter Famine and the siege of Leningrad, demonstrate that in-utero exposure to food insecurity may lead to a "thrifty phenotype"—a phenomenon where poor nutrition in early life permanently alters glucose-insulin metabolism, increasing the risk of chronic diseases like type-2 diabetes and metabolic syndrome later in life — in offspring^{15, 19}. Food insecurity also contributes to obesity, anaemia, and other diet-sensitive health outcomes, particularly among vulnerable populations like the elderly²⁰. In conjunction with physical health outcomes, food insecurity has a profound impact on mental health. Studies in the United States have shown that household food insecurity is associated with developmental risk, reduced academic performance, and adverse effects on social skills development in children³. Food insecurity also increases the risk of depression, anxiety, and stress, with potential mechanisms that includes stress, social isolation, and poor diet quality⁴.

Food Insecurity and Depression

In South Africa, studies have shown a correlation between food insecurity and depression, with food-insecure households being the most affected^{21, 22}. “Depression is a complex mental health disorder characterized by persistent feelings of sadness, loss of interest, and disturbed sleep and/or appetite”⁴. Globally, an estimated 322 million individuals are affected, with a 4.6% prevalence rate in South Africa in 2010^{4, 23}. Depression is a multidimensional disorder, assessed using measures like the Patient Health Questionnaire (PHQ-9), Beck Depression Inventory II (BDI-II), and Centre for Epidemiological Studies Depression Scale (CES-D)²³. These tools evaluate cognitive, affective, and somatic symptoms, enabling accurate diagnoses and appropriate interventions²⁴.

Empirical evidence reveals a strong link between food insecurity and depression, underscoring the vulnerability of food insecure individuals with poor mental health, anxiety, and stress^{2, 25}. Living with food insecurity can lead to depression due to the associated anxiety and uncertainty, and engaging in socially unacceptable behaviours to obtain food, which can perpetuate isolation, disempowerment, and shame⁴. The National Income Dynamics Study-Coronavirus Rapid Mobile (NIDS) revealed that in March 2021, over a third of the population (35%) faced food affordability crises, increasing their likelihood of exhibiting depressive symptoms²².

The link between food insecurity and depression is intricate and influenced by various factors. Several explanations have been suggested to account for this connection, including stress, anxiety, and social isolation. Stress is a key mediator in the relationship between food insecurity and depression. Food insecurity can lead to chronic stress, which can alter cortisol levels, brain 5-HT function, and ultimately contribute to depressive symptoms²⁶. Additionally, food insecurity can lead to physiological distress, including increased insulin resistance, elevated lipid, and C-reactive protein levels, and altered adipokine levels, which can further exacerbate depressive symptoms²⁷.

Anxiety is another mechanism underlying the relationship between food insecurity and depression. Food insecure individuals may experience anxiety due to the uncertainty and worry about accessing nutritious food, resulting in pervasive and distressing worry about common, everyday concerns²⁸. Social isolation is also a critical factor in the relationship between food insecurity and depression. The emotional burden of food insecurity – including shame, guilt, and anxiety – can lead to social isolation, decreased community involvement, and reduced social interactions²⁸. Depressive symptoms can also mediate the relationship between food insecurity and cognitive function. Research indicates that depressive symptoms acts as a mediator in the link between food insecurity and impaired cognitive function, implying that food insecurity can contribute to depression, which negatively impacts cognitive performance²⁹.

Social Support and Community Resources

“Social support is a complex and multifaceted concept that includes physical and instrumental assistance, attitude transmission, resource and information sharing, and emotional and psychological support”³⁰. This can be categorized into two main types: responsive support, which addresses specific needs, and ambient support, derived from membership in a social group. This support encompasses five key forms — emotional

support, providing reassurance and empathy; instrumental support, offering financial or material aid; informational support, delivering advice and guidance; companionship support, fostering a sense of belonging; and validation support, providing feedback on social status³¹. Researchers use standardized measures like the Social Support Questionnaire (SSQ) and the Multidimensional Scale of Perceived Social Support (MSPSS) to assess these types of social support³².

Community resources available to address food insecurity and depression include food assistance programs (e.g., food banks, meal delivery services) and mental health services (e.g., counselling, therapy, support groups)³³. Food distribution systems involving Non-Governmental Organizations (NGOs) and international aid agencies can rapidly and efficiently deliver assistance to vulnerable communities³⁴. Charitable Food Assistance Programs (CFAPs) like food banks and food rescue organizations play a crucial role in addressing food insecurity³³. However, despite these resources, a significant proportion of individuals struggling with mental health conditions remain untreated or lack access to necessary care³⁵.

Social capital, comprising social networks and community cohesion, significantly influences access to community resources for addressing food insecurity and depression³⁶. Through social networks, individuals can access various support structures, including informational guidance, material assistance, and emotional sustenance³⁶. Social capital plays a vital role in enhancing food security by fostering collaborative relationships among community members throughout the food supply chain⁹. By developing trust, empathy, and a shared sense of social norms, social capital enables individuals to work together towards common objectives, ultimately ensuring more reliable access to nutritious food⁹.

Role of Social Support and Community Resources in Reducing Depression

Research conducted in the United States reveals that social support and community resources can reduce depression and alleviate its symptoms³⁷. Theoretical frameworks, such as social capital theory, suggest that social support and community resources can increase access to various forms of support during times of need³⁶. Social capital can facilitate access to food, childcare, and other forms of support, which can reduce stress and alleviate symptoms of depression^{36, 38}. Additionally, social relationships and support can act independently or modify the effects of life events, such as food insecurity, to reduce the risk of depression³⁸.

Cultural and contextual factors, such as socioeconomic status, ethnicity, and urban vs. rural settings, can moderate the relationship between social support, community resources, and depression³⁹. Research has shown, for example, that the impact of social network factors varies by sex, with food-insecure men potentially feeling embarrassed among wealthy friends, leading to further symptoms of depression⁴⁰. Additionally, social support has been found to be unequally distributed among social classes, with lower social classes having less access to social support, increasing their vulnerability to health complaints⁸.

2.1 RESEARCH GAP

Despite the established link between food insecurity and depression, and the recognized importance of social support and community resources in mitigating mental health impacts, a research gap exists regarding the impact and effectiveness of these resources in reducing depression among individuals experiencing food insecurity in South Africa.

3. PROBLEM STATEMENT

Approximately, 1 in 9 households suffer from food insecurity, which poses significant risks to mental well-being, particularly depression¹. The complex connection between food insecurity, social support, and community resources on depression is not yet fully understood, hindering the development of effective interventions. Therefore, there is a need to investigate the effectiveness of social support and community resources in reducing depression among individuals experiencing food insecurity in South Africa, to inform strategies that lessen the mental health impacts of food insecurity.

4. AIMS & OBJECTIVES

4.1 RESEARCH QUESTION

What role do social support and community resources play in reducing depression related to food insecurity?

4.2 AIMS

To investigate the effectiveness of social support and community resources in reducing depression among individuals experiencing food insecurity in South Africa.

4.3 OBJECTIVES

- Determine the prevalence rate of depression among individuals experiencing food insecurity in South Africa.
- Investigate the relationship between food insecurity and depression in South Africa.
- Examine the mediating role of social support and community resources in reducing depression among individuals experiencing food insecurity.

5. METHODOLOGY

5.1 Study Design

This study employs a secondary data analytical cross-sectional design. Utilizing data from the 2018 – 2022 General Household Survey, this study captures a snapshot of the relationship between food insecurity, social support, community resources, and depression among households in South Africa at a specific point in time.

5.2 Study Setting

This study utilizes secondary data from the 2018 – 2022 GHS conducted by Stats SA, covering all nine provinces — [Gauteng (GP), Western Cape (WP), Kwa-Zulu Natal (KZN), Mpumalanga (MP), Eastern Cape (EC), Free State (FS), Limpopo (L), Northern Cape (NC) and North West (NW)] — of South Africa, home to approximately 60.6 million people. The study focuses on households and individuals experiencing food insecurity. The diverse geographic and socio-economic conditions across the provinces provide valuable insights into the complex relationships between food insecurity, social support, community resources, and depression outcomes.

5.3 Study Population

The study will consist of individuals who participated in the 2018 – 2022 GHS.

5.4 Sampling Method

The GHS employed a two-stage stratified design, wherein Primary Sampling Units (PSUs) were selected using Probability Proportional to Size (PPS) sampling, followed by systematic sampling of Dwelling Units (DUs) within chosen PSUs. The sample was subsequently grouped by province and then further categorized within each province using data from Census 2011, with primary stratification based on population characteristics such as age, gender, and income, and secondary stratification based on geographic characteristics such as urban/rural location and region.

The GHS was conducted in all nine provinces from 2018 to 2022. In 2018, 20 908 households were engaged, while 19 351 households participated in 2022. Due to the COVID-19 pandemic, data collection methods were adapted for 2020 and 2021. Specifically, Computer-Assisted Telephone Interviews (CATI) replaced face-to-face interviews, utilizing the 2019 sample with operational telephone numbers. In 2020, this approach yielded responses from 8 896 household, while in 2021, 9 629 households participating. Adjustments were made for non-contacts, out-of-scope dwellings, and weighting processes to ensure data accuracy (detailed in the Technical Notes). Throughout the survey period, trained survey officers, supported by provincial and district coordinators, and quality assurors, ensured data quality across provinces.

5.5 Sample Size Determination

The 2018 – 2022 GHS draws upon the 2013 Master Sample, developed using data from Stats SA's 2011 Census. The Master Sample is structured to ensure representation at both provincial and sub-provincial levels, with further stratification at metropolitan/non-metropolitan levels. The 2011 Census divided the country into 103 576 Enumeration Areas (EAs). The census EAs, coupled with additional data, served as the fundamental units for constructing the PSUs of the Master Sample. This was due to their comprehensive national coverage and possession of essential information for stratification and PSU creation. The Master Sample comprises 3 324 PSUs, yielding an anticipated sample of around 33 000 DUs. In reflection, the current Master Sample has expanded by 8.0% compared to the previous 2008 version, increasing from 3 080 to 3 324 PSUs.

Weighting

The sampling weights were designed to expand the sample data to reflect the entire South African population. The weights were constructed by considering several key factors, including design weights, PSU adjustments, sampling frame exclusions, non-response, weight trimming, and calibration to known population benchmarks from Stats SA. Household estimates based on the UN headship ratio methodology were used to weight the household data, ensuring accurate representation.

Sampling for this study

This study will utilize Stats SA's GHS (2018-2022) secondary data, encompassing 275 902 participants from 78 433 households. The dataset combines survey results from 2018 (20 908 households, 71 137 individuals), 2019 (19 649 households, 68 986 individuals), 2020 (8 896 households, 34 370 individuals), 2021 (9 629 households, 35 265 individuals), and 2022 (19 351 households, 66 144 individuals). The sample included individuals from all nine provinces in South Africa, providing a comprehensive representation of the population experiencing food insecurity. By utilizing this secondary data, the study leveraged a robust sampling design and a large, nationally representative sample, allowing for reliable estimates and insightful analyses of the relationships between food insecurity, social support, community resources, and depression outcomes in South African households.

5.6 Data Collection

This study involves secondary data collection using the GHS from Stats SA. The national statistical agency, has been conducting the GHS for over twenty years, making it one of its longest-running surveys. The GHS is a continuous survey that collects data on various key themes, such as basic services, healthcare, education, agriculture, and more.

The survey adopted a multifaceted approach, covering a wide range of topics that explored various aspects of participants' lives. The questionnaire encompassed demographic characteristics, household composition, education, health, social grants, economic activities, housing, communication, transport, welfare, food security, livelihoods, and mortality. With depression, participants were questioned about their depression diagnosis, overall health status, and healthcare consultations. Interviewers carefully documented household information, responses, and other pertinent data to ensure a comprehensive and accurate documentation process.

5.7 Variables

Dependent variables

The binary outcome variable for this study will be depression.

Independent variables

The independent variables to be considered will encompass age, gender, education level, employment status, insufficient food for children, insufficient food for adult, skipped meals, main source of income.

Mediator variables

Social support and community resources refer to the networks and services that provide emotional, informational, and tangible assistance to individuals in need.

Social support will include marital status, household size, household income.

Community resources will encompass means of transport to nearest health facility, provision of food, Participated in Government/municipal job creation programme.

Table 1: Variables Table

Variable Name	Variable Type	Definition	Scale
Age Group	Categorical	Is a segment of a population that is approximately the same age group.	00 – 14 years 15 – 29 years 30 – 44 years 45 – 59 years 60 – 74 years 75+ years
Gender	Binary	A characteristic that defines humans as female or male.	Female/Male
Education Level	Categorical	Is a subdivisions of formal learning covering childhood education, primary, secondary, and tertiary education.	No schooling Foundation education Secondary education Higher education
Marital Status	Categorical	Indicates whether an individual is married, single, or in a legally recognized partnership or union.	Legally married Living together Divorced Legally married, but separated Widowed Cohabiting, but unmarried Single and have never been married

Depression	Binary	A psychological condition described by prolonged emotional distress, reduced satisfaction, and feelings of hopelessness, significantly impacting daily life and wellbeing.	Yes/No
Household Size	Categorical	The total count of persons residing in a single dwelling unit.	1 – 3 4 – 6 7 – 9 10+
Household Income	Categorical	The total monthly income earned by all members of a household.	R0 – R9 999 R10 000 – R19 999 R20 000 – R29 999 R30 000 – R39 999 R40 000+
Employment Status	Binary	An individual's current employment situation.	Employed/Unemployed
Insufficient food for children	Categorical	The frequency of not having enough food for children in the household.	Never Seldom Sometimes Often Always
Insufficient food for adult	Categorical	The frequency of not having enough food for adults in the household.	Never Seldom Sometimes Often Always
Skipped meals	Binary	The frequency of missing meals because of food unavailability or financial constraints.	Yes/No
Main source of income	Categorical	The primary source of income for an individual or household.	Salary/Wage Pensions Grants
Means of transport to nearest health facility	Categorical	The primary mode of transportation used to travel to the nearest health facility.	Walking Taxi Bus Train Own transport Bicycle
Provision of food	Categorical	Receipt of free or subsidized meals or food assistance provided by the school to learners, aiming to support their nutritional needs and improve learning outcomes.	No Yes, everyday Yes, a few times a week Yes, sometimes

Participated in Government or municipal job creation programme	Binary	Involvement in a government-initiated or municipally-funded program designed to create temporary or permanent employment opportunities, skills training, or income-generating activities for individuals, often targeting unemployed or underemployed populations.	Yes/No
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5.8 Measurement Instrument Design

The study's data will be obtained through the 2018 – 2022 Stats SA demographic survey, which employed a structured questionnaire carefully designed and validated from the Stats SA household survey. Researchers adapted the questionnaire to align with the study's context and objectives and then administered it using in-person interviews and CATI – with skilled interviewers – to ensure accurate and reliable data.

5.9 HOW TO OVERCOME MEASUREMENT ERROR (QUALITY ASSURANCE/DATA MANAGEMENT)?

To optimize research impact and reliability, this study's Data Management Plan (DMP) ensures data is created, stored, and shared in a secure and accessible format, facilitating easy collaboration. A two-stage sample design was implemented to ensure representativeness in the primary survey, featuring stratified selection of primary sampling units by size, systematic sampling of dwelling units, and province-level stratification based on geographic and demographic characteristics from the 2011 Census.

For this secondary study, I will obtain the original dataset from Stats SA and rigorously clean and prepare it using Stata 18, addressing missing values and inconsistencies. New variables will be created, and existing data will be recorded to align with research objectives. To ensure data quality, I will thoroughly review the original study's technical manuals, methodology, sampling techniques, and analysis weights. Questionnaires and interview protocols will be scrutinized to understand data collection processes.

To mitigate measurement error and enhance reliability, Stats SA has standardized training protocols that are implemented for interviewers and data managers, covering data management best practices, ethical considerations, and legal compliance. Quality assurance measures include; regular audits and data validation checks, secure data storage and sharing protocols, data validation techniques such as cross-checks and consistency reviews , ongoing data quality monitoring and control processes.

5.10 DATA ANALYSIS PLAN

Data will be analysed using StataBE version 18.

Objective 1: Determine the prevalence rate of depression among individuals experiencing food insecurity in South Africa.

To summarize demographic characteristics and food insecurity status, descriptive statistics will be utilized. The prevalence of depression will be assessed using the depression measure included in the GHS dataset, and the prevalence rate of depression among individuals experiencing food insecurity will be determined, stratified by demographic characteristics. The results will be presented utilizing frequency distribution tables to illustrate key findings. Categorical data will be summarized as charts, whereas continuous data will be described using measures of central tendency (means).

Objective 2: Investigate the relationship between food insecurity and depression in South Africa.

Bivariate analysis will be employed to identify any association between food insecurity and depression. A multivariate logistic regression will then be utilized to control for any potential confounding variables and determine the adjusted odds ratio of depression among individuals experiencing food insecurity.

Objective 3: Examine the mediating role of social support and community resources in reducing depression among individuals experiencing food insecurity.

I will employ mediation analysis to determine the extent to which social support and community resources mediate the association between food insecurity and depression. This will enable us to identify potential pathways through which social support and community resources can help alleviate the negative psychological impacts of food insecurity.

6. ETHICAL CONSIDERATIONS

This study is conducted in accordance with ethical principles, ensuring participant confidentiality and safeguarding rights. Since this research utilizes secondary data from the GHS conducted by Stats SA, no direct participant contacts or data collection is involved. The original data collection process by Stats SA ensured informed consent, anonymity, and confidentiality. This study's analysis of existing data does not pose risks to participants. All data analysis will be conducted using anonymized and aggregated data, ensuring participants' privacy. Results from this study will contribute to the development of research-informed interventions aimed at food insecurity and depression, ultimately promoting the well-being of vulnerable populations in South Africa.

7. TIMELINE

Table 2: Timetable of activities

Activity	17 Nov 2024	10 June 2025	22 June 2025	30 June 2025	10 July 2025	25 July 2025
Research protocol development						
Write Project draft						
Submit draft for approval						
Editing and corrections						
Approval by supervisor						
Submit for evaluation						

8. BUDGETING

Table 3: Research Budget Plan

	Costs (in Rands)			
	0 - 999	1 000 – 1 999	2 000 – 2 999	3 000 – 3 999
Data acquisition costs				
Data analysis tools				
Miscellaneous (printing, conference attendance)				
Database subscriptions				

9. REPORTING OF RESULTS

A comprehensive research report will be submitted to the University of Pretoria's (UP) research database, ensuring accessibility for the academic community. I will also present findings at UP organized conferences, symposiums, and workshops to engage with individuals who are more experienced. Furthermore, I will share study outcomes through student-friendly journals, social media, and other organizations (including the data provider – Stats SA) to promote public awareness and understanding.

10. CONCLUSION

This study aims to explore the role of social support and community resources in reducing depression among food-insecure households in South Africa. By addressing this knowledge gap, the research hopes to inform targeted interventions and improve the well-being of vulnerable populations.

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